Spatial Relationships

Which body parts are used?

- **All body parts** listed in your project's bp_names.csv are used for pairwise distances and convex hulls.
- **Center point** features use the body part defined in your code as the "center" (default in your constants: **bodypart14**).
- **X vs Y movement** uses **every** body part individually, then also gives you a summed score.

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1) X vs Y movement (per-frame bias: horizontal vs vertical)

These columns indicate whether motion is more **horizontal** (X) or **vertical** (Y) per frame.

- **Movement_{i}_X_relative_2_Y**
  For each body part (indexed by its position in your list), we compute
  (Δx) − (Δy) between consecutive frames.
    - Positive → more X than Y movement in that frame
    - Negative → more Y than X
    - **Units: pixels/frame** (note: this method does **not** convert to mm)
- **Movement_X_axis_relative_to_Y_axis**
  Sum of all Movement_{i}_X_relative_2_Y across body parts in the frame.
    - Quick whole-animal bias score (X vs Y)
    - **Units: pixels/frame**

Typical examples
Movement_0_X_relative_2_Y, Movement_1_X_relative_2_Y, … and the aggregate Movement_X_axis_relative_to_Y_axis.

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2) X vs Y rolling windows (trend over time)

- **Movement_X_axis_relative_to_Y_axis_mean_{W}**
  Mean of the aggregate bias over the last **W frames**.
  Units: pixels/frame

- **Movement_X_axis_relative_to_Y_axis_sum_{W}**
  Sum of the aggregate bias over the last **W frames**.
  Units: pixels

Example
Movement_X_axis_relative_to_Y_axis_mean_150,
Movement_X_axis_relative_to_Y_axis_sum_150
(~5 s windows if 30 fps)

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3) Distances between body parts (all pairs)

For every **pair** of body parts (combinations of your bp_names):

- **Distance_{bpA}_{bpB}**
  Euclidean distance between body parts bpA and bpB per frame, converted to mm.
  Units: mm

Rolling stats per window **W** for each pair:

- **{Distance}_{mean_{W}}** → average distance (mm) over W frames
- **{Distance}_{std_{W}}** → standard deviation (mm) over W frames
- **{Distance}_{skew_{W}}** → skewness (unitless)
- **{Distance}_{kurtosis_{W}}** → kurtosis (unitless)

Examples
Distance_bodypart2_bodypart16,
Distance_bodypart2_bodypart16_mean_150,
Distance_bodypart2_bodypart16_std_150,
Distance_bodypart2_bodypart16_skew_150,
Distance_bodypart2_bodypart16_kurtosis_150

A few rows may have -1 for skew/kurtosis if the window is too short or numerically unstable.

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4) Convex hull of the skeleton (whole-fish "spread")

We build the convex hull from **all body parts' (x, y)** in a frame and compute its **perimeter**.

- **Convex_hull**
  Convex hull **perimeter** of all body points (converted to mm).
  Units: mm (perimeter length)

Rolling stats per window **W**:

- **Convex_hull_mean_{W}_window** → mean (mm)
- **Convex_hull_std_{W}_window** → std (mm)
- **Convex_hull_min_{W}_window**, **Convex_hull_max_{W}_window** → min/max (mm)
- **Absolute_diff_min_max_convex_hull_{W}_window** → range = max − min (mm)
- **Convex_hull_skew_{W}**, **Convex_hull_kurtosis_{W}** → shape (unitless)

Examples
Convex_hull, Convex_hull_mean_150_window, Convex_hull_std_150_window,
Convex_hull_min_150_window, Convex_hull_max_150_window,
Absolute_diff_min_max_convex_hull_150_window,
Convex_hull_skew_150, Convex_hull_kurtosis_150

Read it as a **compactness / spread** cue: larger values mean the body parts are more spread out (e.g., fins extended).

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5) Spatial occupancy (center point drift & variability)

Uses your **center body part** (default constant: bodypart14).

- **Center_displacement**
  Distance from the **start position** of the center point to the current frame's center.
  Units: mm

- **Center_x_variance**, **Center_y_variance**
  Rolling variance of the center's **x** and **y** across a short window (~fps/2 frames).
  Units: pixels^2 (coordinates aren't converted in the current code)

Example
If fps = 30, the variance window is ~15 frames.

If you need variance in **mm^2**, scale your x/y before variance, or divide these by (px_per_mm)^2.
